

Wavelet-Based Reconstruction of Saturn Ring Bending Waves from Cassini Observations

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ABSTRACT

We present a modular analysis pipeline for extracting and modeling bending waves in Saturn’s rings using radial brightness profiles derived from Cassini ISS observations, and apply it to the full available image sets of two observing sequences: six images of revolution 114 and twelve images of revolution 115. The workflow consists of (i) photometric normalization using fixed reference bands to define a stable background intensity, (ii) continuous wavelet transforms to estimate the local radial wavenumber as a function of radius, with automatic gap-filling, harmonic correction, and per-wave edge detection, and (iii) forward modeling of the wave signal using both least-squares and neural-network approaches. Wave phases are constructed by integrating the local wavenumber, and amplitudes are inferred through sinusoidal reconstructions that incorporate the viewing geometry via the effective opening angle B_{eff} . We report extracted wavenumber fields, reconstructed amplitudes, and model–data comparisons for one representative image per revolution (rev 114.34 and rev 115.109), a structured comparison between the least-squares and neural-network approaches, and a comparison of reconstruction consistency across all images within each revolution, including an identified geometric trend of inferred amplitude with B_{eff} .^{1,2}

1. INTRODUCTION

Vertical bending waves in Saturn’s rings arise when resonant perturbations force coherent oscillations of ring material out of the ring plane.^{3–5} Unlike density waves, which primarily modulate surface density, bending waves are observed through changes in the local surface slope, and therefore provide a direct probe of the vertical structure of the disk and its dynamical response to external forcing.³ In reflected-light observations, this sensitivity to slope produces characteristic brightness modulations whose wavelength and amplitude evolve with radius as the wave propagates away from its resonance.

The data analyzed in this work were acquired by the Imaging Science Subsystem (ISS) aboard the *Cassini* spacecraft.⁶ The ISS camera provides two-dimensional images of Saturn’s rings with sufficient spatial resolution to resolve fine-scale wave structure in the Cassini Division.⁶ We consider two observation sequences here: 114_R1EQXSHADOW and 115_R1LAZSCANLIT. For each sequence, multiple ISS images were obtained under well-characterized viewing geometries, enabling the reconstruction of radial brightness variations across the wave-bearing region. The full analysis pipeline is applied to every available image in both sequences (six images for rev 114, twelve for rev 115); for this paper, we adopt the 34th image of revolution 114 (rev 114.34) and the 109th image of revolution 115 (rev 115.109) as detailed case studies, and separately summarize trends across the full image sets in Sec. 4.4.

Figure 1 and Fig. 2 show representative ISS images from rev 114.34 and rev 115.109, respectively.⁶ In each case, the Cassini Division is clearly visible, along with subtle brightness variations associated with bending-wave structure. These images constitute the fundamental observational input to the analysis presented in this paper.

For quantitative analysis, the two-dimensional ISS images were reduced to one-dimensional radial brightness profiles by azimuthally averaging each radii over the entire image. This procedure suppresses small-scale azimuthal structure while preserving the radial oscillations induced by bending waves.³ The resulting profiles form the basis for all subsequent analysis in this work. Details of the image calibration, averaging geometry, and profile construction are deferred to a later section.

Extracting physically meaningful parameters from these radial profiles presents several challenges. The local wavenumber of a bending wave varies continuously with radius, rendering global Fourier techniques inadequate.

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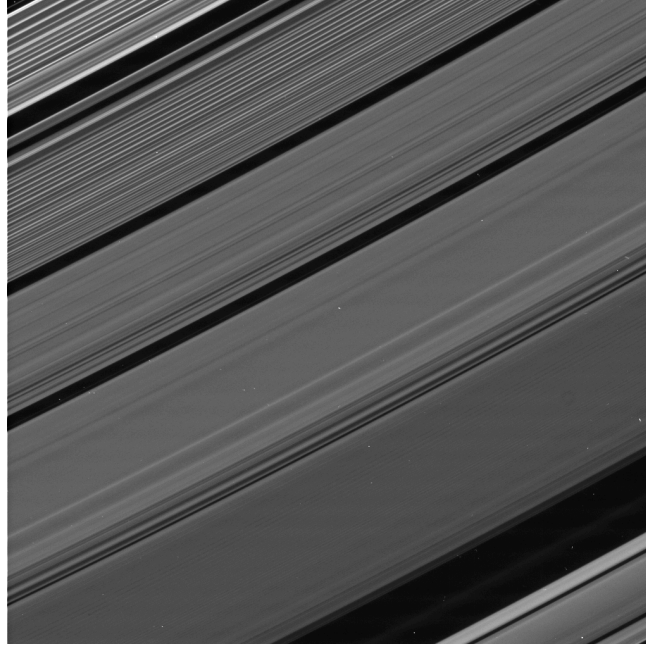


Figure 1. *Cassini* ISS image rev 114.34 from EQXSHADOW showing the Cassini Division. We label each band from top left to bottom right Bands 1-5, with the Huygens gap at the very top left, and Laplace gap near the bottom right.

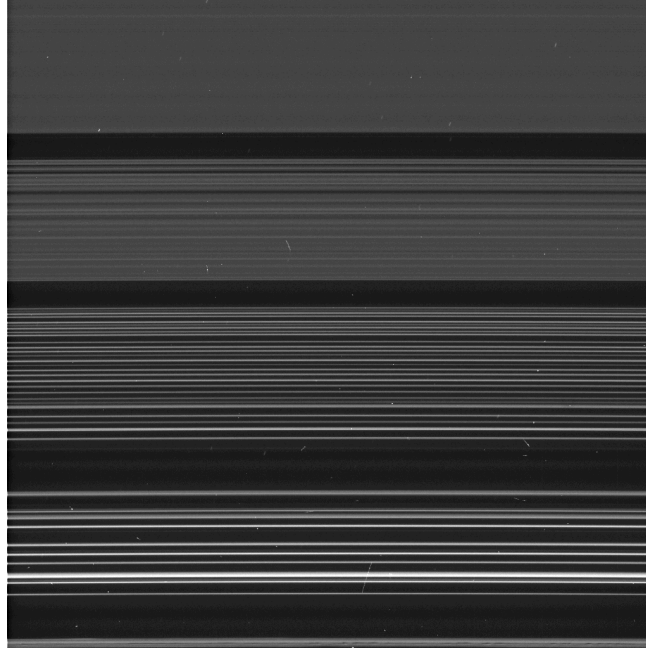


Figure 2. *Cassini* ISS image rev 115.109 from AZSCANLIT showing the Cassini Division. We label each band (correspondingly) from bottom to top Bands 1-4, with the Huygens gap at the very bottom.

In addition, the observed signal is modulated by viewing geometry through the effective opening angle B_{eff} . These considerations motivate the development of a multistage analysis framework that combines time–frequency methods, explicit physical modeling, and data-driven approaches.^{1,2}

In this paper, we present such a framework and apply it consistently to two independent ISS observing sequences. Continuous wavelet transforms are used to estimate the local radial wavenumber as a function of radius,^{1,2} phases are constructed by direct integration, and wave amplitudes are inferred through a sliding-

window least-squares reconstruction based on a linearized forward model. A neural-network model is introduced as a complementary surrogate that preserves the same physical structure while allowing greater flexibility in how amplitude information is inferred. Together, these components enable a detailed, internally consistent comparison of bending-wave structure across multiple Cassini observing sequences.

2. DATA AND OBSERVATIONS

2.1 Cassini ISS image data

All observational inputs used in this work originate from the Imaging Science Subsystem (ISS) aboard the *Cassini* spacecraft.⁶ The ISS camera acquired two-dimensional reflected-light images of Saturn’s rings with sufficient spatial resolution to resolve fine-scale structure within the Cassini Division.⁶ For the observing sequences considered here, the effective radial resolution of the ISS images is approximately 1.2 km per pixel, setting the intrinsic spatial scale of the measurements.

The present analysis focuses on two observing sequences:

- 114_R1EQXSHADOW, comprising ISS images obtained by a pair of radial scans during revolution 114. As such, this scan only contains six images that cover the inner Cassini Division.
- 115_R1AZSCANLIT, comprising ISS images obtained by a pair of azimuthal scans at constant radii during revolution 115.

For each revolution, multiple images were available and were treated consistently within a single analysis configuration. However, as rev 114 is a radial scan, only a limited number of images (six) were taken of the Cassini Division. Rev 115 sequence has a larger number of useful images in this regard. The raw ISS images themselves are retained and referenced in this work, but they are not analyzed directly in two dimensions. Instead, they serve as the source material from which one-dimensional radial brightness profiles are constructed.

These particular sequences were chosen as they are the closest high-resolution ring images to the Saturnian equinox in August of 2009 (Revolution 116). This results in a quite low B_{eff} , of around half a degree. This low angle creates maximum contrast in the reflected intensity of any bending waves. In addition, this low angle allows us to treat the rings as opaque in our modeling, opposed to their true sparse nature. For reference, rev 114.34 has $B_{\text{eff}} = -0.48^\circ$, and rev 115.109 has $B_{\text{eff}} = 0.29^\circ$.

2.2 Conversion from images to radial profiles

The radial brightness scans were extracted from the Flux-calibrated ISS images by M. El Moutamid and P. Nicholson, using software written for the ISS team by M.S. Tiscareno and B.E. Harris.⁷

Each ISS image was mapped into ring-plane coordinates using the appropriate spacecraft geometry and then reduced to a one-dimensional radial profile by azimuthal averaging over a selected annular sector of the Cassini Division.⁶ This averaging procedure suppresses localized azimuthal structure while preserving coherent radial oscillations associated with bending waves.³

The resulting profiles represent azimuthally averaged $I/F(r)$ measurements as a function of radius. Because the original image resolution is ~ 1.2 km, the radial sampling of the profiles is of comparable scale, ensuring that the wavelet analysis probes physically resolved wavelengths. In particular, the typical wavelength of the bending waves in these images is 10 to 15 kilometers, as such a resolution of ~ 1.2 km resolves them quite well. These profiles constitute the primary data products used throughout the wavelet, least-squares, and neural-network analyses described in subsequent sections.

In this paper, we refer to the brightness of any pixel as I/F , representing the observed intensity (radiance) I , divided by the incident solar flux (irradiance) F . We approximate the rings in these images to be a Lambert surface with zero absorption. Thus $I/F = \cos i$, independent of the observer’s location, where i is the solar incidence angle.

2.3 Radial domain and normalization

The analysis is restricted to a fixed radial interval within the Cassini Division that contains the bending-wave signal of interest. All radial masks, trimming of edge regions, and exclusion of contaminated intervals are specified explicitly in the analysis configuration file (`config.py`) and applied identically to both rev 114.34 and rev 115.109.

Prior to wave analysis, each radial profile is normalized by a reference brightness level to correct for azimuthal and/or temporal variations in brightness due to varying phase or emission angle. This reference brightness level I_0/F is derived from two disjoint radial bands outside the main wave region. Normalization occurs simply by dividing the entire region by this value. * After normalization, a mean value for each wave region is subtracted to center individual waves at 0 for ease of use in modeling, particularly for the neural network. Thus, the quantity analyzed is the residual brightness $\frac{I/F}{I_0/F} - (I/F)_{avg}$, which isolates the oscillatory component associated with the bending wave.³ In practice, for a good normalization value I_0/F , we expect $(I/F)_{avg}$ to be quite close to unity.

2.4 Profile sampling and consistency across revolutions

All radial profiles are sampled on a uniform radial grid, with a sampling interval specified in `config.py` and inherited directly by the wavelet analysis.^{1,2} The same sampling, normalization, and masking procedures are applied to both observing sequences, ensuring that differences observed between rev 114.34 and rev 115.109 arise from the data themselves rather than from methodological inconsistencies.

The normalized and trimmed radial profiles form the sole observational input to the subsequent stages of analysis: continuous wavelet transforms for local wavenumber estimation,^{1,2} phase construction by integration, least-squares amplitude reconstruction, and neural-network modeling.

2.5 Note on image-level analysis

Although this paper focuses on one-dimensional radial profiles, the underlying two-dimensional ISS images are preserved and available.⁶ Future extensions of this work may incorporate image-level or azimuthally resolved analyses. For clarity, however, all results presented here are derived exclusively from azimuthally averaged radial profiles constructed from ISS images with an intrinsic spatial resolution of approximately 1.2 km.

3. METHODOLOGY

This section describes the full analysis pipeline implemented in the accompanying code directory. The workflow operates on one-dimensional, normalized, azimuthally averaged radial profiles $I/F(r)$ constructed from Cassini ISS images (Sec. 2). Two images are processed using the same sequence of steps: (i) wavelet-domain estimation of a local wavenumber field and ridge extraction, (ii) least-squares reconstruction of a physically motivated sinusoidal model with two distinct regression stages, and (iii) a physics-informed neural-network that predicts amplitude components while retaining the same forward model and phase construction.¹⁻³

Throughout, radii are expressed in kilometers, and wavenumbers are represented either as cycles/km (for grid definition and visualization) or radians/km (for phase integration and forward modeling). When explicit numerical values are defined in `config.py`, they are stated below. In particular, global normalization is anchored to two fixed reference regions which correspond to bands three and four respectively,

$$\text{GLOBAL_BAND_1} = (118650, 118850) \text{ km}, \quad \text{GLOBAL_BAND_2} = (119000, 119200) \text{ km},$$

and analysis is performed on wave-specific radial windows: wave 1, $r \in [117960, 118070]$ km; wave 2, a nominal window $r \in [118350, 118530]$ km whose true extent is refined automatically by the ridge-detection stage (Sec. 3.2); wave 5A, the inner portion of wave 5, $r \in [119460, 119560]$ km; and wave 5, the full extent including the outer envelope, with a nominal outer bound of 119700 km that is likewise refined automatically. Wave 5A and wave 5 share a local baseline band used for renormalization, $\text{WAVE5A_BASELINE_BAND} = (119450, 119724)$ km.

*These normalization bands are fixed in radius and defined in `config.py`.

3.1 Pipeline architecture

The methodology above is implemented as a small, modular Python codebase organized around a single shared configuration and a linear per-image processing chain. This subsection summarizes the codebase structure so that the results in Sec. 4 can be traced back to the specific stage that produced them.

Configuration. `config.py` centralizes every image-independent constant used throughout the pipeline (wave windows, normalization bands, CWT grid parameters, ridge masks, NN loss weights) as well as a per-image table of effective opening angles `B_EFF_TABLE`, keyed by `"{rev}.{img}"`. A single function, `config.reconfigure(rev, img_num)`, repoints all image-dependent paths (input profile, output directory, plot directory) and geometry values at a new image, so that the same downstream code operates identically on any image without modification.

Stage 1 — CWT and ridge extraction (`analysis/cwt_pipeline.py`, using the wavelet engine in `helpers/cwt.py`). Loads the pre-normalized radial profile for the configured image, computes the wavelet power spectrum and raw ridge over the bandwidth grid, applies the corrections described in Sec. 3.2 (harmonic removal, gap filling, per-wave edge detection and smoothing), and writes a single `cwt_ridge_result.npz` file per image containing the raw and smoothed ridge, the auto-detected wave 2/wave 5 bounds, and the normalized brightness profile. This file is the sole input to every later stage.

Stage 2 — Least-squares reconstruction (`analysis/ls_waves.py`, using `helpers/physics.py`). Loads the Stage-1 ridge output, and for each requested wave builds the phase by trapezoidal integration and fits the sliding-window sinusoidal model of Sec. 4.2, saving one amplitude `.npz` file and diagnostic plots per wave.

Stage 3 — Neural-network reconstruction (`analysis/nn_waves.py`, using the model defined in `helpers/nn_model.py`). Loads the same Stage-1 ridge output (and, where available, the Stage-2 LS amplitude for side-by-side comparison plots), constructs the five-feature input described in Sec. 4.3, trains `ContextAmpCSNet`, and saves the resulting amplitude `.npz` file and diagnostic plots per wave.

Batch drivers (`scripts/run_analysis.py`, `scripts/run_all.py`). `run_analysis.py` runs Stages 1–3 in sequence for the single image currently set in `config.py`, automatically checking each wave’s point coverage against a minimum-points threshold before attempting a fit and skipping waves that fall below it. `run_all.py` performs the same sequence for every profile file found in `data_profiles/`, which is how the full six-image (rev 114) and twelve-image (rev 115) result sets used in Sec. 4.4 were generated.

Cross-image summaries (`scripts/plot_amplitudes.py`, `scripts/plot_vs_beff.py`). Once every image in a revolution has been processed, these scripts aggregate the per-image amplitude `.npz` files into overlay and stacked comparison figures (Sec. 4.4) and into amplitude-versus- B_{eff} trend plots, respectively.

Because every stage after Stage 1 consumes only the contents of `cwt_ridge_result.npz` and the shared `config` module, the LS and NN stages are fully decoupled from one another and from the wavelet engine — a property exploited directly in the comparison of Sec. 5.

3.2 Wavelet transform and ridge extraction

The first stage of the analysis estimates the local radial wavenumber of each bending wave as a function of radius. Because the wavelength varies continuously across the wave region, this is performed using a continuous wavelet transform rather than a global Fourier analysis.

Each normalized radial brightness profile is transformed using a complex Morlet wavelet, evaluated over a dense grid of wavenumbers spanning 0.02–0.30 cycles km^{-1} . For reference, this corresponds to wavelengths between 3.3km and 50km. The Morlet wavelet provides simultaneous localization in radius and wavenumber, allowing the dominant oscillatory scale to be tracked as a function of position. Multiple wavelet bandwidths (`BANDWIDTH_GRID = {0.8, 1.0, 1.2, 1.5, 2.0}`) are explored, and the bandwidth that maximizes ridge coherence is selected automatically, to ensure robustness of the ridge identification.

The wavelet power spectrum $P(r, k)$ is computed as the squared magnitude of the transform coefficients. At each radius, a raw local wavenumber $k_{\text{raw}}(r)$ is estimated by identifying the wavenumber at which the power is maximized. To suppress spurious maxima caused by overlapping structures or wave harmonics, restricted regions of (r, k) space are excluded from consideration when selecting the ridge, implemented as a list of $(r_{\text{min}}, r_{\text{max}}, k_{\text{min}}, k_{\text{max}})$ boxes (`RIDGE_MASKS`) in which the power is set to $-\infty$ prior to the arg-max search.

Two such masks are applied: one suppressing a spurious high-power band in wave 2 ($r \in [118497, 118590]$ km, $k \in [0.05, 0.095]$ cycles/km), and one suppressing the first harmonic overtone across the full wave 5 extent ($r \in [119450, 119720]$ km, $k > 1/(2\pi)$ cycles/km). These regions were chosen upon analysis of the power spectrum to enforce continuity of the physically relevant ridge across each wave-bearing region.

3.2.1 Ridge correction and per-wave edge detection

Even after masking, the raw ridge $k_{\text{raw}}(r)$ retains two classes of defect that a single global smoothing pass cannot fully address: residual harmonic contamination in wave 5, and a short data gap in wave 2 where the wavelet ridge loses lock. Both are corrected explicitly before smoothing.

For wave 5, any raw samples with $k_{\text{raw}} \geq 0.9$ rad/km (just below the 1.0 rad/km mask ceiling) are treated as residual harmonic and replaced by linear interpolation over the surrounding valid samples, producing a harmonic-corrected raw ridge.

For wave 2, a focused radial sub-window ($r \in [118495, 118535]$ km) is searched for a coupled drop-rise pair in the sample-to-sample derivative Δk : the first sample with $\Delta k < -0.35$ rad/km marks the start of the gap, and the first subsequent sample with $\Delta k > +0.35$ rad/km marks its end. A gap is only registered when both a drop and a rise are found; the enclosed samples are then replaced by linear interpolation using the raw ridge on either side of the gap as anchors.

With the raw ridge corrected, the hard radial extent of each wave (its start and end) is located from the corrected raw ridge itself, *before* smoothing: the start is the radius of the single largest positive derivative ($\arg \max \Delta k$, the sharpest background→wave transition) and the end is the radius of the single largest negative derivative ($\arg \min \Delta k$, the sharpest wave→background transition), each restricted to the wave’s configured search window. This derivative-based approach is more robust than a fixed-threshold crossing, which can land inside a gradual ramp rather than at the true edge. The resulting bounds (`wave2_rmin/rmax`, `wave5_rmin/rmax`) are recorded in the saved ridge output and used, in place of the nominal `config.py` windows, by all downstream least-squares and neural-network stages for waves 2 and 5.

Only the samples strictly inside each detected wave segment are then smoothed, using a 21-sample boxcar filter with edge-padding drawn from the segment’s own boundary values rather than from the full profile. Smoothing each wave in isolation this way prevents background-level values just outside a wave from bleeding into the smoothed ridge at the wave’s edges, which would otherwise bias the earliest and latest phase samples used in reconstruction. Waves 1 and 5A are smoothed the same way, over their fixed `config.py` windows (no gap or harmonic correction is needed for these regions). The resulting per-wave smoothed ridge $k(r)$ defines the local wavenumber field used for all subsequent modeling.

3.3 Least-squares modeling

The least-squares (LS) approach reconstructs the bending-wave signal by fitting a local sinusoidal model to the normalized brightness residuals. Using the ridge-derived wavenumber $k(r)$, the wave phase is constructed by direct integration,

$$\psi(r) = \int_{\text{Wave}} k(r) dr,$$

Where wave radii are selected in accordance with linear trends in wave number as found in the ridge extraction. Within a local radial neighborhood, the residual brightness is modeled as

$$I/F_{\text{model}}(r) = I_0/F [1 - (a(r) \cos \psi(r) + b(r) \sin \psi(r))].$$

where $a(r)$ and $b(r)$ are treated as slowly varying coefficients. Note that I_0/F was divided out during previous steps in practice. These coefficients are estimated via linear least-squares regression over a sliding window with length comparable to the local wavelength.

The fitted slope amplitudes,

$$\sqrt{a^2(r) + b^2(r)} = \frac{k(r) * A_V}{\cot B_{\text{eff}}},$$

represent the projected surface-slope signal in the absence of shadowing. The corresponding vertical displacement amplitude is obtained as

$$A_V(r) = \frac{1}{|k(r)|} \frac{\sqrt{a^2(r) + b^2(r)}}{\cot B_{\text{eff}}},$$

where B_{eff} is the effective opening angle of the observation.

3.4 Neural-network modeling

The neural-network (NN) approach is designed as a physics-informed surrogate for the least-squares reconstruction. The NN does not replace the physical forward model; instead, it replaces the local sliding-window estimation of the sinusoidal amplitude components by learning smooth, radius-dependent amplitude fields that are coupled to the same ridge-derived phase construction.

3.4.1 Inputs and predicted quantities

For each wave window, the NN operates on a one-dimensional radial sequence. At each radius, the input feature vector includes (all normalized to zero): the radius r , the ridge-estimated local wavenumber $k(r)$, the local wavenumber gradient dk/dr , an inverse-wavenumber term $1/(|k| + \epsilon)$ to encode local wavelength scale, and the geometric factor $\cot B_{\text{eff}}$. The NN predicts two outputs at each radius, interpreted as cosine and sine amplitude components,

$$A_c(r), \quad A_s(r).$$

Note that these amplitudes are vertical amplitudes of the bending wave, opposed to the dimensionless slope amplitudes used the least squares fit above. The physically relevant vertical displacement amplitude is then

$$A_V(r) = \sqrt{A_c^2(r) + A_s^2(r)}.$$

3.4.2 Network architecture

The NN is a lightweight one-dimensional convolutional sequence model with two convolutional layers followed by two fully connected layers. The input has dimension 5, one per feature, one per radius sample, and the hidden width is 64. The architecture is:

- **Conv1:** 1D convolution, $5 \rightarrow 64$ channels, kernel size 5, with padding chosen to preserve sequence length.
- **Conv2:** 1D convolution, $64 \rightarrow 64$ channels, kernel size 5, with padding chosen to preserve sequence length.
- **FC1:** linear layer, $64 \rightarrow 64$.
- **FC2:** linear layer, $64 \rightarrow 2$, producing (A_c, A_s) .

Rectified linear unit (ReLU) nonlinearities are applied after each of the first three layers. The final layer is linear so that the amplitude components may take positive or negative values, consistent with phase-dependent decomposition into cosine and sine components.

3.4.3 Shared phase construction and forward physics

As in the least-squares approach, the ridge-derived wavenumber defines the phase through trapezoidal integration,

$$\psi(r) = \int_{\text{Wave}} k(r) dr,$$

and the NN outputs are inserted into the same sinusoidal forward model. The modeled brightness profile takes the normalized form

$$I/F_{\text{model}}(r) = 1 - \cot B_{\text{eff}} k(r) (A_c(r) \cos \psi(r) + A_s(r) \sin \psi(r)).$$

Note that in practice, we re-normalize with the $I/F_{0,\text{band}}$ average found in wave band. With a perfect global normalization, this value should be quite close to unity. From this model, the NN is constrained to explain the data using the same oscillatory basis and geometric scaling as the LS formulation; only the inference mechanism for (A_c, A_s) differs.

3.4.4 Training objective and regularization

Training minimizes the mean-squared error between the modeled and observed profiles over the wave window,

$$\mathcal{L}_{\text{data}} = \left\langle \left(I/F_{\text{model}} - I/F_{\text{data}} \right)^2 \right\rangle,$$

with additional smoothness regularization applied directly to the predicted amplitude components $A_c(r)$ and $A_s(r)$, rather than to the derived envelope $A_V(r)$. Penalties on the first and second radial differences of each component are included to discourage un-physical high-frequency structure in the recovered amplitude:

$$\mathcal{L}_{\text{smooth}} = \langle (\Delta A_c)^2 \rangle + \langle (\Delta A_s)^2 \rangle, \quad \mathcal{L}_{\text{curv}} = \langle (\Delta^2 A_c)^2 \rangle + \langle (\Delta^2 A_s)^2 \rangle,$$

and the total loss is taken as

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_1 \mathcal{L}_{\text{smooth}} + \lambda_2 \mathcal{L}_{\text{curv}},$$

with fixed weights $\lambda_1 = 1000$ and $\lambda_2 = 100$. Because the raw data term is $\mathcal{O}(1)$ while $\Delta A_c, \Delta A_s$ are amplitude-scale quantities (of order 10^{-2} – 10^{-1} in the units used internally), these larger weights are needed for the smoothness terms to meaningfully compete with the data term during training; the network is trained for 2000 epochs on waves 1 and 2 and 4000 epochs on waves 5 and 5A, using the Adam optimizer with a learning rate of 10^{-3} .

Because the NN shares the same phase construction and forward physics as the LS method, comparisons between the two approaches isolate differences in how amplitude information is inferred (local analytic regression versus learned contextual mapping), rather than differences in physical assumptions.

4. RESULTS

This section presents the results of the wavelet-based wavenumber extraction, least-squares (LS) reconstruction, and neural-network (NN) modeling applied to Cassini ISS radial profiles. For each image, a single continuous wavelet transform is computed and used for all wave regions present in that data set. For each modeled wave, the model–data comparison and the inferred amplitude profile are presented together as vertically stacked composite figures, constructed from separate code-generated plots.

4.1 Wavelet power and ridge extraction

For each revolution, a single continuous wavelet transform is computed over the full radial domain encompassing all analyzed waves. Figures 3, 5, 4, and 6 show the wavelet power spectra and extracted ridge wavenumbers for rev 114.34 and rev 115.109, respectively. In both cases, we identify waves via ridges that follow a smoothly varying wavenumber. This is consistent with inward-propagating bending waves.

The ridge-mask suppression, harmonic correction, gap filling, and per-wave edge detection and smoothing applied to produce this ridge are described in full in Sec. 3.2. The resulting smoothed ridge $k(r)$ serves as the common input to all subsequent LS and NN modeling stages.

4.2 Least-squares reconstruction

4.2.1 Rev 114.34

Wave 1. The LS reconstruction for wave 1 in rev 114.34 yields a coefficient of determination $R^2 = 0.511$ (R^2 value of unity denotes a perfect fit). Figure 7 shows the stacked LS results, with the model–data comparisons above the inferred amplitude profiles. The reconstruction captures the dominant oscillatory structure but systematically underestimates the amplitude and develops a noticeable phase offset toward the outer portion of the wave.

The inferred vertical amplitude has a mean of ~ 24 m and a peak of ~ 54 m, decreasing overall across the wave with notable drops near the center and right edge. These artifacts are mirrored in amplitude dips in the I/F itself. This decrease is not strictly monotonic and includes localized enhancements, indicating sensitivity of the LS estimator to departures from a purely sinusoidal waveform.

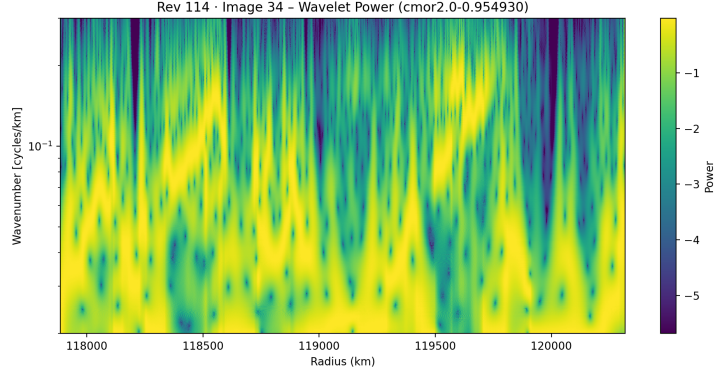


Figure 3. Wavelet power spectrum for rev 114.34.

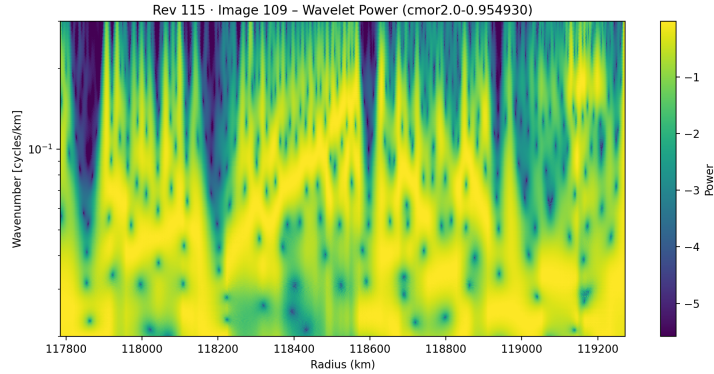


Figure 4. Wavelet power spectrum for rev 115.109.

Wave 2. The LS reconstruction for wave 2 in rev 114.34, evaluated over its auto-detected extent $r \in [118350, 118530]$ km (Sec. 3.2), yields $R^2 = 0.292$ (Fig. 7). The reconstruction again captures the dominant oscillatory structure but systematically underestimates the amplitude and develops a noticeable phase offset.

The inferred vertical amplitude has a mean of ~ 8 m and a peak of ~ 18 m, decreasing overall across the wave region. Similar to the first wave, this demonstrates sensitivity to non sinusoidal waveforms.

Wave 5. For wave 5 in rev 114.34, the LS reconstruction performs more poorly, with $R^2 = 0.109$ over the full auto-detected extent (Fig. 7). The first half of the wave is not well described by a sinusoidal model, resulting in substantial residual structure. The fit slightly improves at larger radii where the signal becomes more nearly sinusoidal. Restricting the fit to the inner wave 5A sub-region alone improves the fit substantially, to $R^2 = 0.744$, indicating that most of the residual structure originates in the outer portion of the wave 5 extent.

The inferred amplitude is of order $A_V \sim 0.4\text{--}0.6$ m (mean) with peaks below 1 m, decreasing with radius, which is plausible. However, this is unlikely to represent physical data given the resolution of our images.

4.2.2 Rev 115.109

Wave 1. For wave 1 in rev 115.109, The LS reconstruction yields $R^2 = 0.485$ (Fig. 8). As in rev 114.34, the model exhibits systematically low amplitude and noticeable phase drift. Unlike rev 114.34, however, the inferred amplitude does not show a clear radial decrease and instead displays a more irregular structure (mean ~ 26 m, peak ~ 54 m). In addition, we see that the model predicts there to be a fourth peak at around $40 + 1.18e5$ km, when none is to be found in the I/F. This could be a result of a smaller peak being shadowed by a larger, previous peak, but no explanation is definitive for this absent crest.

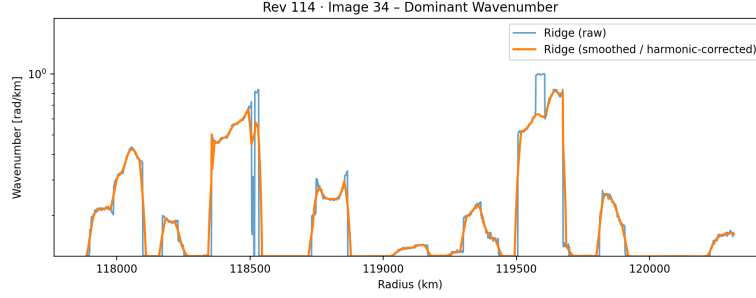


Figure 5. Ridge Trace of Wavelet power spectrum for rev 114.34.

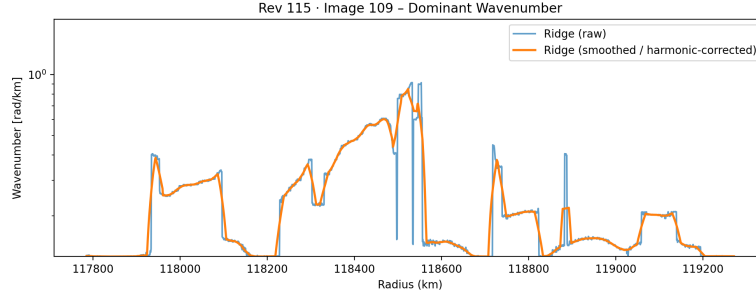


Figure 6. Ridge Trace Wavelet power spectrum for rev 115.109.

Wave 2. For rev 115.109, The LS reconstruction yields $R^2 = 0.393$ (Fig. 8). Similarly, the model exhibits systematically low amplitude (mean ~ 9 m, peak ~ 19 m) and noticeable phase drift. In addition, like rev 114.34, the model is able to capture a jump in amplitude, near the center of the I/F.

4.3 Neural-network reconstructions

4.3.1 Rev 114.34

For rev 114.34, neural-network reconstructions are available for waves 1, 2, and 5. In all cases, the NN provides substantially improved agreement with the observed profiles relative to the LS approach.

Wave 1. The NN reconstruction for wave 1 achieves $R^2 = 0.985$ (Fig. 9, mean $A_V \approx 19$ m, peak ≈ 45 m), a high-quality reconstruction of the observed oscillatory structure that demonstrates the flexibility of the NN framework relative to the corresponding LS fit ($R^2 = 0.511$).

Wave 2. The NN reconstruction for wave 2 achieves $R^2 = 0.925$ (Fig. 9, mean $A_V \approx 8$ m, peak ≈ 20 m), reproducing the oscillatory structure with minimal phase drift. The inferred amplitude decreases smoothly with radius, consistent with expectations for an inward-propagating bending wave.

Wave 5. For the full auto-detected extent of wave 5 in rev 114.34, the NN reconstruction yields $R^2 = 0.917$ (Fig. 9, mean $A_V \approx 0.4$ m, peak ≈ 1.1 m). The fit quality is high across the wave region; however, the inferred amplitude does exhibit some radial decay expected on physical grounds, though more oscillatory than expected. Restricting to the inner wave 5A sub-region gives a comparable $R^2 = 0.869$; see the remark on wave 5A at the end of this subsection.

Wave 5A, the fixed inner sub-region of wave 5 retained for continuity with earlier configurations of this pipeline, is not discussed as a separate case study here: because it is a strict radial subset of the wave 5 region already presented above, its reconstruction is qualitatively similar (NN $R^2 = 0.869$, LS $R^2 = 0.744$) and does not add independent information beyond what the full wave 5 extent already shows. It remains available in the

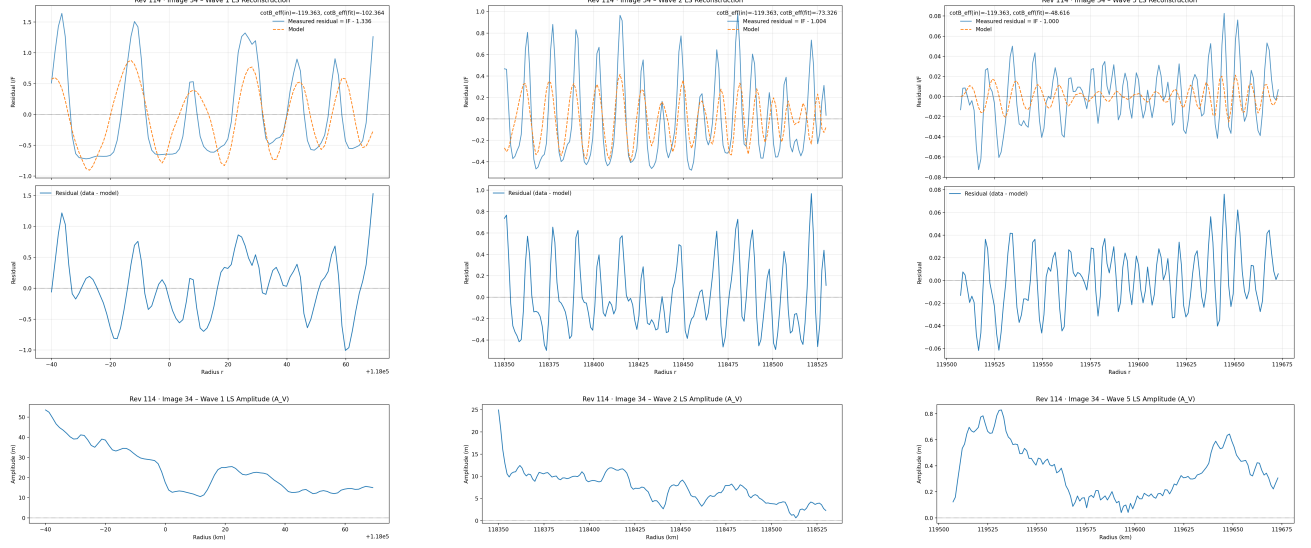


Figure 7. Least-squares reconstructions for rev 114.34. *Left*: wave 1 ($R^2 = 0.511$). *Center*: wave 2 ($R^2 = 0.292$). *Right*: wave 5 ($R^2 = 0.109$). For each wave, the model–data comparison is shown above the inferred amplitude $A_V(r)$.

output directory structure (NN/Wave 5A/, LS/Wave 5A/) for images where the outer wave 5 envelope is not well resolved.

4.3.2 Rev 115.109

For rev 115.109, neural-network reconstructions are available for waves 1 and 2. As in rev 114.34, the NN approach provides substantially improved agreement with the observed radial profiles relative to the least-squares method.

Wave 1. The NN reconstruction for wave 1 achieves $R^2 = 0.992$ (mean $A_V \approx 31$ m, peak ≈ 88 m), reproducing the observed oscillatory structure with high fidelity (Fig. 10). The model–data comparison shows minimal residual structure, but the inferred amplitude varies strongly across the wave region. In fact, due to a lack of coherent period, the network has simply modeled wake peaks as individual spikes in amplitude. The NN result here demonstrates the pitfall of the framework in regimes where a simple local sinusoidal approximation is insufficient.

Wave 2. For wave 2 in rev 115.109, the NN reconstruction again provides a marked improvement over the LS result, achieving $R^2 = 0.903$ (mean $A_V \approx 11$ m, peak ≈ 22 m; Fig. 10). The oscillatory structure is reproduced with minimal phase drift across the full radial extent of the wave. Compared to the LS reconstruction, which exhibits systematically low amplitudes and irregular radial behavior, the NN-inferred amplitude is smoother and more coherent.

Unlike wave 2 in rev 114.34, however, the NN amplitude profile for rev 115.109 does not exhibit a clear monotonic decrease with radius. This behavior may reflect genuine physical differences between the two observing geometries, sensitivity to the smaller (magnitude) effective opening angle ($B_{\text{eff}} = 0.29^\circ$), or residual limitations of the NN regularization. These effects are discussed further in the following section.

Wave 5. A reminder that rev 115.109 does not cover the fifth band, or wave 5.

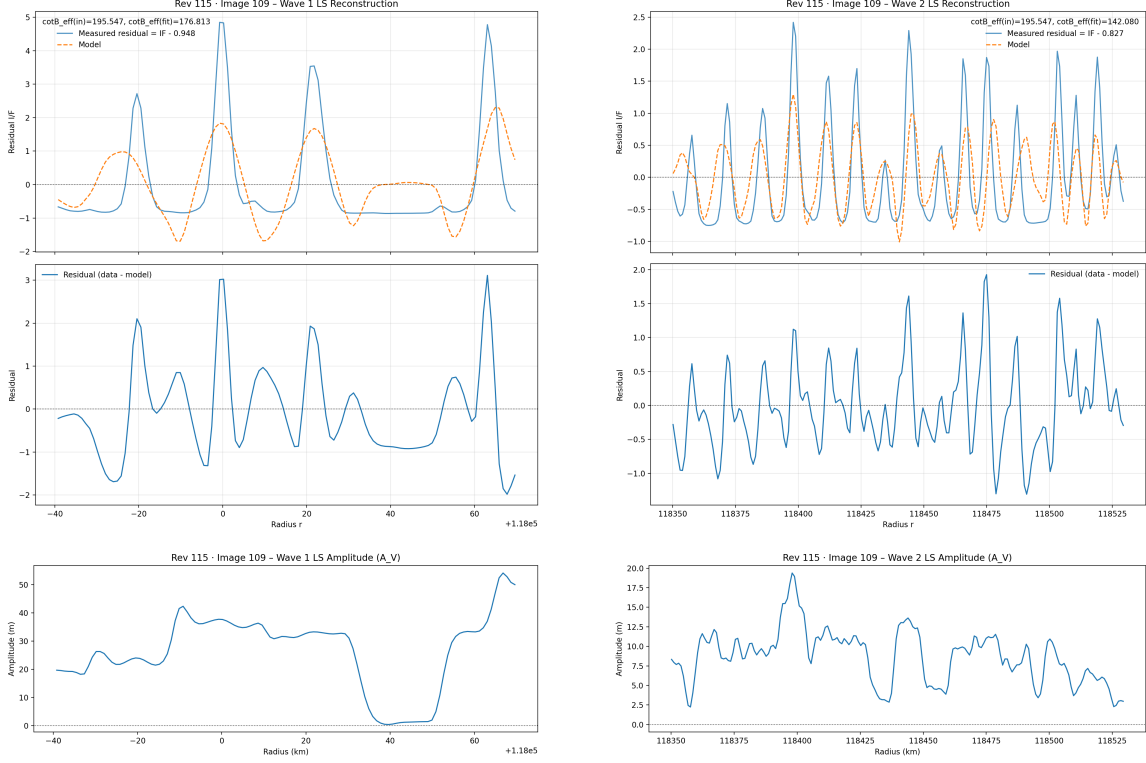


Figure 8. Least-squares reconstructions for rev 115.109. *Left*: wave 1 ($R^2 = 0.485$). *Right*: wave 2 ($R^2 = 0.393$). For each wave, the model–data comparison is shown above the inferred amplitude $A_V(r)$.

4.4 Multi-image comparison

The case studies above (rev 114.34, rev 115.109) fix the pipeline’s behavior on a single image per revolution. Because `scripts/run_all.py` (Sec. 3.1) applies the identical pipeline to every available image, we can additionally ask how consistent the NN-inferred amplitudes are *within* a revolution, across all six rev 114 images (34, 35, 36, 37, 62, 63) and all twelve rev 115 images (109, 120, 130, 140, 150, 160, 170, 180, 190, 200, 210, 217). Not every wave is resolved in every image: wave 1 is unavailable for images 62 and 63 of rev 114, and wave 5/5A are unavailable for all of rev 115 (Sec. 1).

Table 1. NN reconstruction statistics aggregated across all available images per revolution. A_V values are the per-image mean, averaged again across images; ranges give the minimum–maximum of that per-image mean across the image set.

Wave	n_{img}	R^2 (mean, range)	Mean A_V [m] (range)
<i>Rev 114</i>			
Wave 1	4	0.982 (0.971–0.987)	18.7 (17.8–19.3)
Wave 2	6	0.903 (0.891–0.925)	8.2 (7.6–9.7)
Wave 5A	4	0.844 (0.684–0.978)	0.7 (0.6–0.9)
Wave 5	4	0.837 (0.794–0.917)	0.5 (0.4–0.6)
<i>Rev 115</i>			
Wave 1	12	0.989 (0.971–0.994)	40.0 (30.7–53.8)
Wave 2	12	0.938 (0.885–1.000)	15.7 (6.0–27.3)

Figure 11 shows the NN-inferred $A_V(r)$ overlaid across all available images, for wave 1 and wave 2 of each revolution and for wave 5/5A of rev 114. Within rev 114, whose six images share a common effective opening angle ($B_{\text{eff}} = -0.48^\circ$, Sec. 2), reconstruction quality and inferred amplitude are both tightly clustered across images (R^2 within ± 0.02 of the mean for waves 1 and 2), supporting the internal consistency of the pipeline independent of the specific image analyzed.

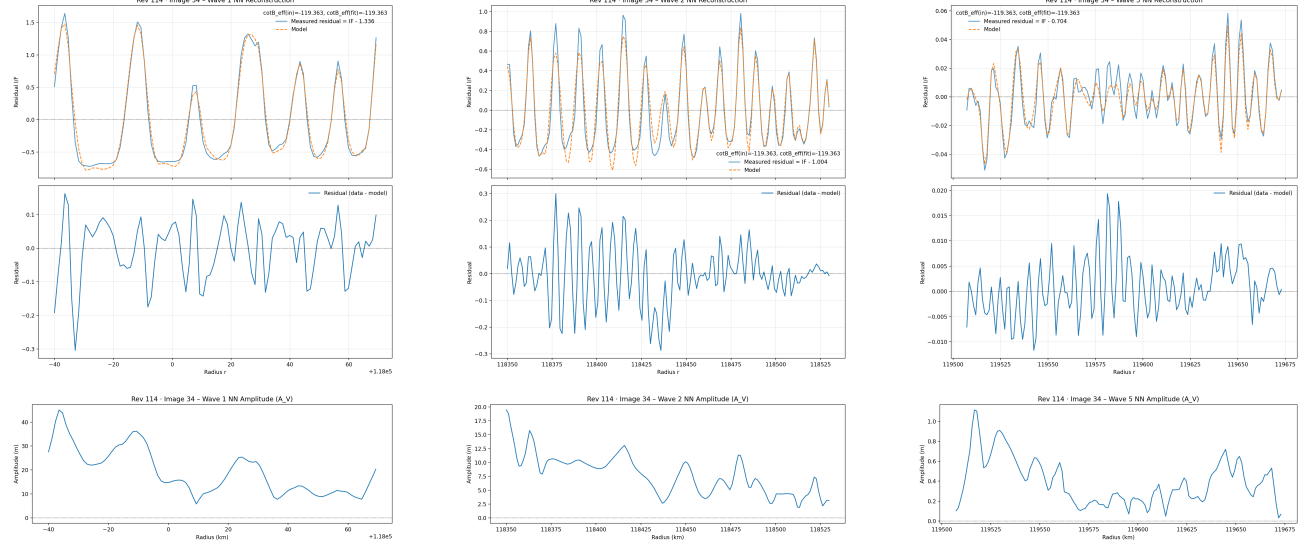


Figure 9. Neural-network reconstructions for rev 114.34. *Left to right:* wave 1 ($R^2 = 0.985$), wave 2 ($R^2 = 0.925$), and wave 5, full auto-detected extent ($R^2 = 0.917$). For each wave, the model–data comparison is shown above the inferred amplitude $A_V(r)$.

Rev 115 is more informative because its twelve images span a five-fold range in B_{eff} (0.293° to 0.714° ; Sec. 2). Across this range, the NN-inferred mean A_V for both wave 1 and wave 2 increases monotonically with B_{eff} — for wave 1, from 30.7 m at $B_{\text{eff}} = 0.293^\circ$ to 53.8 m at $B_{\text{eff}} = 0.714^\circ$. This tracks the decrease of $\cot B_{\text{eff}}$ over the same range (from 195.5 to 80.2) and is consistent with the forward model of Sec. 3, in which $A_V \propto 1/\cot B_{\text{eff}}$ at fixed slope amplitude: any residual, roughly B_{eff} -independent noise floor in the fitted slope amplitude is amplified into a larger inferred A_V as $\cot B_{\text{eff}}$ shrinks. We therefore treat this trend primarily as a geometric artifact of the amplitude-recovery step rather than as direct evidence of a physical dependence of the wave amplitude on B_{eff} , though the two effects are not distinguished by the present analysis and disentangling them is left for future work.

Direct comparison between rev 114.34 and rev 115.109 is restricted to wave 1 and wave 2, present in both data sets. Both LS and NN reconstructions indicate systematic differences in amplitude behavior and phase stability between the two images, plausibly driven in part by the differing effective opening angles, consistent with the within-revolution B_{eff} trend identified above. A more detailed quantitative synthesis of these differences, isolating genuine radial/temporal amplitude evolution from the geometric B_{eff} dependence, is deferred to a dedicated performance-analysis section to be completed later.

5. COMPARISON OF LEAST-SQUARES AND NEURAL-NETWORK APPROACHES

In this section we explicitly compare the least-squares (LS) and neural-network (NN) approaches used to reconstruct bending-wave structure from the Cassini ISS radial profiles. The comparison is made at two levels: (i) an analytical comparison of the modeling assumptions, parameterization, and mathematical structure of the two methods, and (ii) a numerical comparison based on goodness-of-fit metrics and qualitative behavior observed in the reconstructed profiles and inferred amplitudes.

5.1 Analytical comparison

5.1.1 Shared physical framework

Both the LS and NN approaches are built around the same underlying physical forward model. In both cases, the observed brightness residual is represented as

$$I/F(r) = I_0/F[1 - \cot B_{\text{eff}} k(r) [A_c(r) \cos \psi(r) + A_s(r) \sin \psi(r)],$$

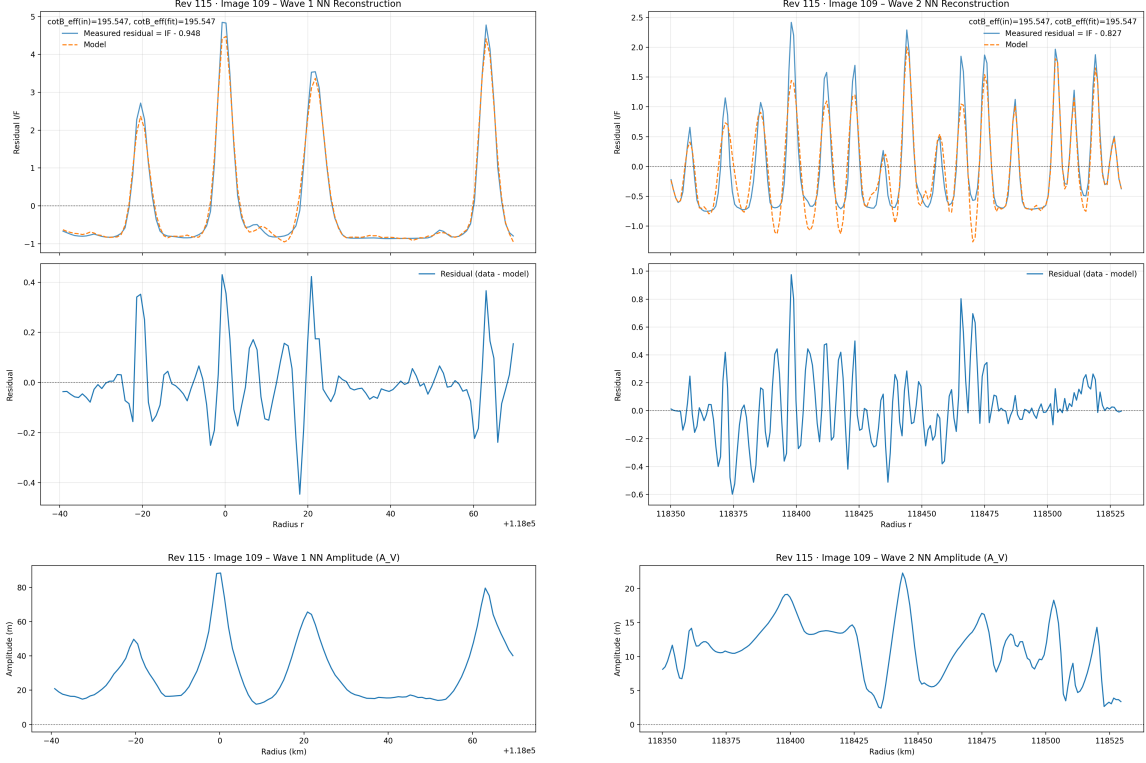


Figure 10. Neural-network reconstructions for rev 115.109. *Left:* wave 1 ($R^2 = 0.992$). *Right:* wave 2 ($R^2 = 0.903$). For each wave, the model–data comparison is shown above the inferred amplitude $A_V(r)$.

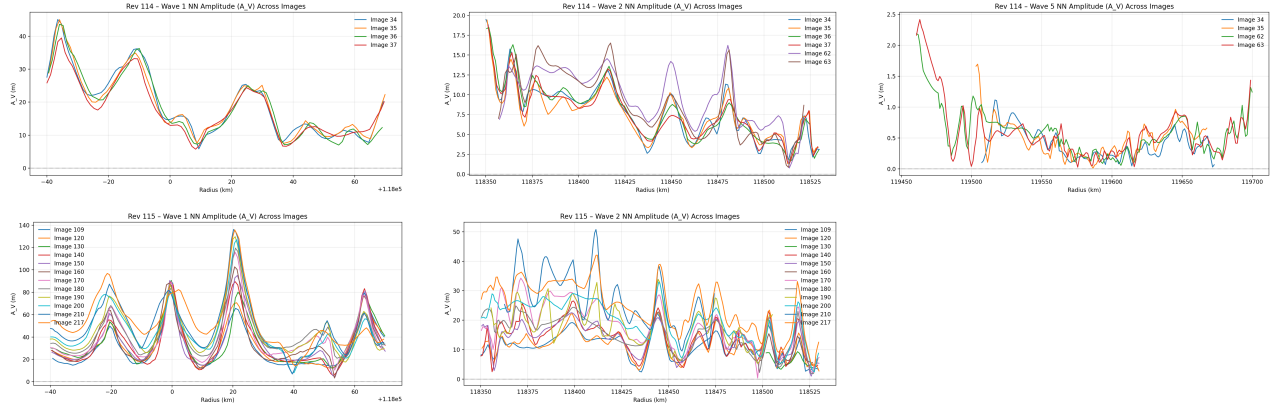


Figure 11. NN-inferred $A_V(r)$ overlaid across all available images within each revolution. *Top row* (rev 114, $n = 4\text{--}6$ images, fixed B_{eff}): wave 1, wave 2, wave 5. *Bottom row* (rev 115, $n = 12$ images, $B_{\text{eff}} \in [0.293^\circ, 0.714^\circ]$): wave 1, wave 2.

where $k(r)$ is the ridge-derived local wavenumber, $\psi(r) = \int k(r) dr$ is the accumulated phase, and B_{eff} is the effective opening angle determined from ISS image metadata (with $B_{\text{eff}} = -0.48^\circ$ for rev 114.34 and $B_{\text{eff}} = 0.29^\circ$ for rev 115.109). In both approaches, the physically relevant vertical displacement amplitude is

$$A_V(r) = \sqrt{A_c(r)^2 + A_s(r)^2}.$$

Crucially, the wavelet-based extraction of $k(r)$ and the trapezoidal phase integration are identical for LS and NN. Differences between the methods therefore arise entirely from how the amplitude components (A_c, A_s) are inferred from the data.

5.1.2 Least-squares parameterization and assumptions

In the LS approach, (A_c, A_s) are assumed to be locally constant within a sliding radial window of length L , chosen to be approximately one local wavelength. Within each window, the model is linear in the parameters, and the coefficients are obtained by solving the normal equations in closed form. This construction imposes several implicit assumptions:

- The signal within each window is well approximated by a single sinusoid, with slowly varying amplitude over the entire fit.
- Deviations from sinusoidality, phase noise, or higher-order structure are absorbed into the residual.
- Amplitude continuity across windows is enforced only indirectly through overlap and smoothing, not through an explicit global constraint.

The LS estimator is therefore highly interpretable and computationally efficient, but it is sensitive to departures from the assumed local model, particularly in regions where the wave shape is distorted or where the signal-to-noise ratio is low.

5.1.3 Neural-network parameterization and assumptions

The NN approach replaces the sliding-window regression with a learned, global mapping from contextual features to $(A_c(r), A_s(r))$. The input features include the local radius, wavenumber, wavenumber gradient, inverse wavenumber, and geometric factor, allowing the model to condition its predictions on the broader wave context.

Unlike the LS method, the NN does not assume that the amplitude is locally constant over a predefined window. Instead, smoothness is enforced through explicit regularization terms in the loss function, penalizing first and second radial derivatives of the predicted components $A_c(r)$ and $A_s(r)$ (Sec. 4.3.4). This allows the NN to accommodate gradual amplitude variation while remaining sensitive to local structure.

Importantly, the NN does not alter the forward physics: phase construction, geometric scaling, and sinusoidal representation are identical to the LS case. The NN therefore acts as a flexible surrogate for the LS amplitude estimator, rather than as an unconstrained black-box regressor.

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